Helix OTA — Continuation

Revision: 4 **Last modified:** 2026-06-22T08:25:00Z

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1. Current state

Field	Value
HEAD	`17cbd47a` `fix(push): portable mkdir lock + honest exit in push_all.sh (§11.4.67/§11.4.1)`
Phase	Stable / release-readiness — all autonomous work GREEN; remaining items hardware/operator-gated
Terminal goal	Fully validated Helix OTA control plane driving real Android A/B updates end-to-end (protocol round-trip → payload apply → slot switch → rollback) on emulated + physical targets

Latest session (2026-06-22) — "do everything workable" sweep

All autonomous items GREEN with real captured evidence (no bluffs): - Test sweep GREEN (server 11/11 + 4 Go submodules 3/3, `-race` 0); **gofmt fixed** (13 files); pre-build gates PASS; docs_chain in-sync. - **Local QEMU A/B OTA 6/6 GREEN** (smoke/boot/slot-switch/rollback/OTA-apply) — evidence `docs/qa/20260622T07*`. Fixed a real §11.4.115 RED-mode polarity isolation bug in `ab_rauc_verity.sh` (RED×2+GREEN×2 deterministic). - **10 server-feature recordings regenerated**, vision-verified (203 HTTP assertions) — `docs/qa/20260622-server-recordings-regen/`; Status.md §11.4.153 reconciled to durable paths. - **§11.4.166 Semgrep**: propagated §11.4.160-166 into CLAUDE/AGENTS/GEMINI; tokenless local scan wired into `commit_all.sh`; constitution pin → `09d8940` (adds `docs/semgrep/TOKEN_SETUP.md`); 7 findings triaged → **0 remaining** (TLS MinVersion fix + cited suppressions). - Submodule push parity closed (`ota-update-engine-bridge`, `ota-android-agent`); GitFlic "blocker" disproven; §11.4.30 transient `qa-results/` untracked+ignored (232 files). - nezha AVD boot proven; real Android A/B = honest **operator-attended SKIP** (Cuttlefish unprovisioned — needs sudo+reboot+fetch_cvd).

Operator action items: (1) Semgrep `SEMGREP_APP_TOKEN` — follow `constitution/docs/semgrep/TOKEN_SETUP.md` (`semgrep login`) to silence the MCP hook (optional; tokenless gate already compliant); (2) provision Cuttlefish on nezha to unblock real Android A/B; (3) RK3588 board for OTA-004/F55/F56.

Known robustness gap (tracked, NOT yet fixed — §11.4.6/§11.4.123): `scripts/boot_android_emulator.sh` — the remote qemu/AVD on nezha stays tied to the launching SSH session and exits gracefully if that session ends mid-boot-wait (the nohup does not fully detach the remote process; the final SSH-tunnel/attestation step isn't reached on interrupt). Boot itself is PROVEN (`emulator-5554`, API 36, `boot_completed=1`, evidence `qa-results/20260622T071848Z-nezha-android-ab/`). Fix candidate: wrap the remote launch in `setsid nohup … </dev/null >log 2>&1 &` (or a `systemd-run --user --scope` on nezha) for true detachment — deferred because on-target persistence verification (re-boot + interrupt test, leaves remote state) is operator-attended; a blind change to the working boot path is forbidden per §11.4.1 without rock-solid proof.

Active items

ID	Title	Status	Type
OTA-003	Emulator Tier-2 — real Android A/B (update_engine/AVB/dm-verity auto-rollback)	In testing	Task
OTA-004	Emulator Tier-3 — real RK3588 / Orange Pi 5 Max vendor HAL, U-Boot slot-switch, dm-verity on real partitions	Operator-blocked	Task

Recently closed items

ID	Title	Status
OTA-021	HelixTrack bidirectional sync verification	Completed
OTA-020	Database migration test	Fixed
OTA-019	Build resource stats tracker	Completed
OTA-018	ApplyPort CLI + slot manager + Ed25519 verifier	Implemented
OTA-017	U-Boot corrupt-slot auto-rollback via bootcount	Implemented
OTA-016	RAUC dd-apply to inactive slot with dm-verity	Implemented
OTA-015	A/B slot switch via U-Boot BOOT_ORDER	Implemented

Full issue details: `docs/Issues.md` · `docs/Fixed.md` · `docs/Issues_Summary.md` · `docs/Fixed_Summary.md`.

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2. Server tests — all GREEN

All 13 testable packages pass (11 internal packages + chaos + stress suites):

```
ok  internal/api
ok  internal/config
ok  internal/device
ok  internal/deviceemu
ok  internal/fabric
ok  internal/health
ok  internal/rollout
ok  internal/store
ok  internal/transport
ok  tests/chaos
ok  tests/stress
```

No-test-file packages (build-only): `cmd/applyport`, `cmd/ota-device-emu`, `cmd/ota-server`, `tools/loadtest`.

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3. Emulator status

Property	Value
Host	`nezha.local` — Linux x86_64, 62 GB RAM, KVM
Image	Android API 36 (Android 16) — `CZ_API36_Phone`
Transport	ADB over SSH tunnel (reachable at `emulator-5554` from nezha)
ADB local	No Android devices connected locally (emulator is remote via SSH tunnel)
LD_LIBRARY_PATH	`/home/milosvasic/.local/lib`
Cuttlefish Tier-2	Pending AOSP guest images

The HelixTrack API is accessible from the emulator via SSH tunnel. The CZ_API36_Phone image provides the standard Android 16 AOSP stack on an x86_64 KVM host.

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4. What's running

- **Main stream:** Emulator Tier-2 validation (remote on nezha.local)
- **Background agents:** None currently dispatched
- **Recording directory:** `$HOME/Downloads` per §11.4.158(D) (default; no project-level override)

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5. Next actions (priority-ordered)

1. **OTA-003 — Emulator Tier-2 inline testing.** Verify the Android emulator on nezha.local can register with the control plane, receive an OTA payload, apply it via `update_engine`, A/B slot-switch, and auto-rollback on corruption. Drive end-to-end through the real user-equivalent path per §11.4.143.
2. **OTA-004 — Hardware unblock.** When a physical RK3588 / Orange Pi 5 Max board becomes reachable over ADB/SSH, flash and validate Tier-3 (vendor HAL, U-Boot slot-switch, real-partition dm-verity).
3. **Feature-coverage video recording.** Produce §11.4.153 mandatory per-feature real-use videos confirming every server endpoint, every emulator tier, and every submodule works — with §11.4.159 window-scoped MP4 capture + §11.4.160 vision verification.
4. **Standing regression-guard suite (§11.4.135).** Ensure every closed OTA-NNN item has its §11.4.115 polarity-switch regression test registered in the suite.

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6. Binding constraints

- **Anti-bluff (§11.4 / §107):** Every PASS carries captured physical evidence per §11.4.5 / §11.4.69 / §11.4.107. Metadata-only / config-only / absence-of-error / grep-without-runtime PASS are all forbidden.
- **No-force-push (§11.4.113):** `git push --force`, `--force-with-lease`, `+<ref>` are STRICTLY FORBIDDEN. Always merge onto latest `main` and push fast-forward-only.
- **Commit via `commit_all.sh`:** All changes use the project's canonical commit wrapper. Direct `git add`/`git commit`/`git push` are never used.
- **Four-layer fix-verification (§11.4.108):** SOURCE → ARTIFACT → RUNTIME-ON-CLEAN-TARGET → USER-VISIBLE. Runtime signature is the definition of done.
- **Independent review (§11.4.142 / §11.4.125):** Every change passes an independent code-review agent before build. Review iterates to zero-finding GO per §11.4.134.
- **Multi-upstream push (§2.1):** Every commit fans out to all four upstreams (GitHub + GitLab + GitFlic + GitVerse).
- **Rootless containers (§11.4.161):** All containerized workloads use Podman in rootless mode via `vasic-digital/containers` submodule.
- **No remote CI (§11.4.156):** All CI/CD automation is DISABLED. Enforcement is through local git hooks + `pre_build_verification.sh`.

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7. Feature tracking

Feature inventory and per-row status: `docs/features/Status.md` (Rev 11, 2026-06-21). Summary companion: `docs/features/Status_Summary.md`.

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8. Fresh session start

```
git fetch --all --prune --tags
```

Then read this file (`docs/CONTINUATION.md`) and `docs/Issues.md` for the full active-item context. The single highest-priority next action is **Emulator Tier-2 inline testing on nezha.local** (OTA-003) — drive the Android emulator through the full OTA lifecycle against the control plane.